

# MATH 189 HW 5

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## Question 1

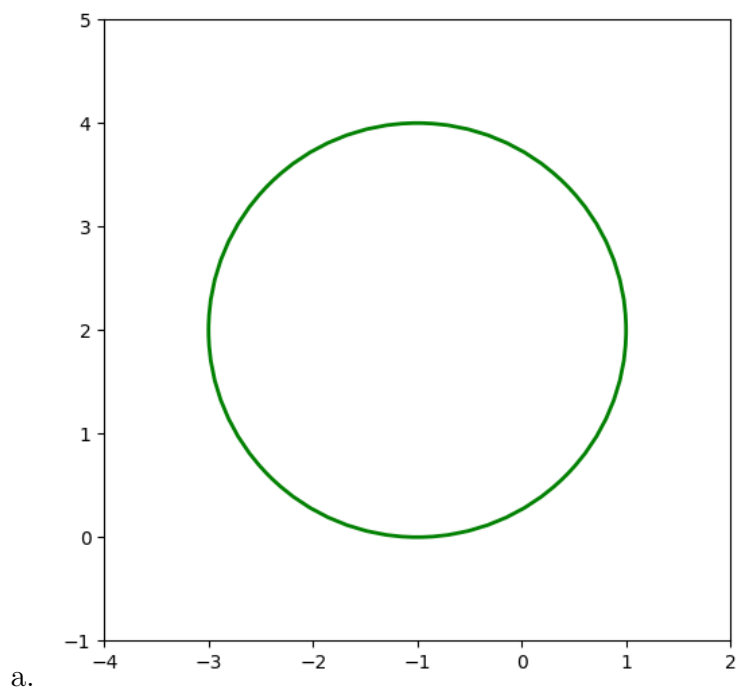
Since boosting takes linear combinations of depth-one trees, which are univariate functions, we get the sum of univariate functions, which is just an additive model.

## Question 2

We split each cell by finding the predictor  $X_j$  and cutpoint  $s$  that minimizes

$$\sum_{i: X_{ij} < s} (Y_i - \hat{Y}_{R_1})^2 + \sum_{i: X_{ij} \geq s} (Y_i - \hat{Y}_{R_2})^2$$

### Question 3

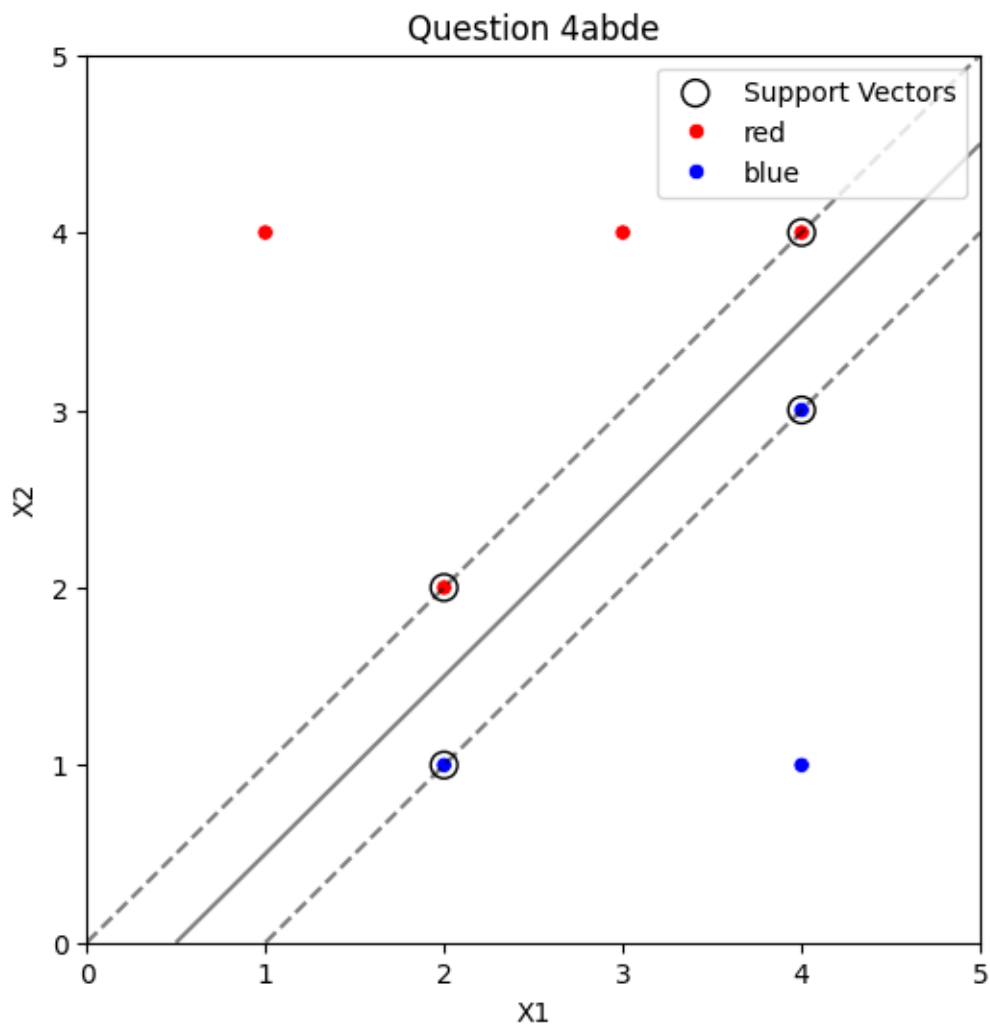


- b. The set of points for which  $(1 + X_1)^2 + (2 - X_2)^2 > 4$  are the points outside the circle, whereas the set of points for which  $(1 + X_1)^2 + (2 - X_2)^2 \leq 4$  are the points on and inside the circle.

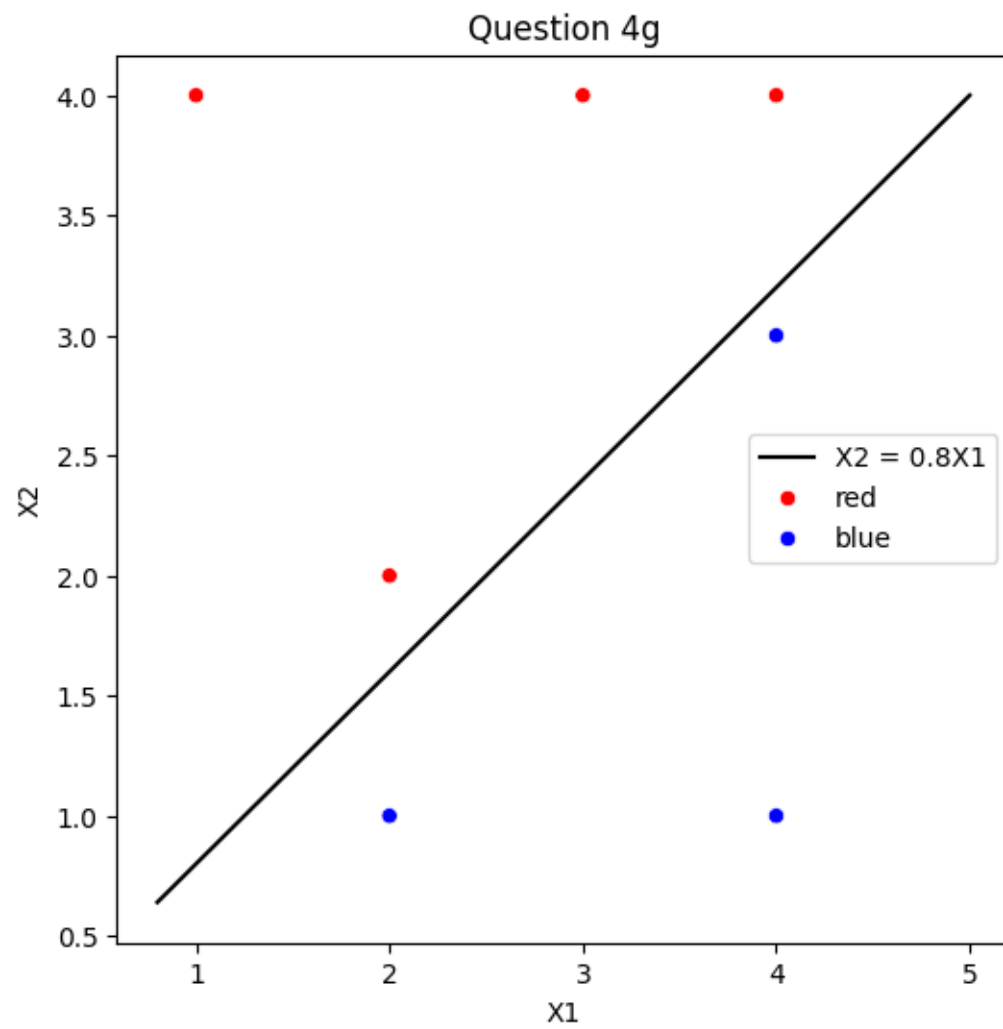
| Points    | $(1 + X_1)^2 + (2 - X_2)^2$ | Class |
|-----------|-----------------------------|-------|
| (0,0)     | 5                           | Blue  |
| c. (-1,1) | 1                           | Red   |
| (2,2)     | 9                           | Blue  |
| (3,8)     | 52                          | Blue  |

- d. If we expand the equation for the decision boundary to  $5 + 2X_1 - 4X_2 + X_1^2 + X_2^2 = 4$ , we get an equation that is linear in terms of  $X_1, X_2, X_1^2, X_2^2$

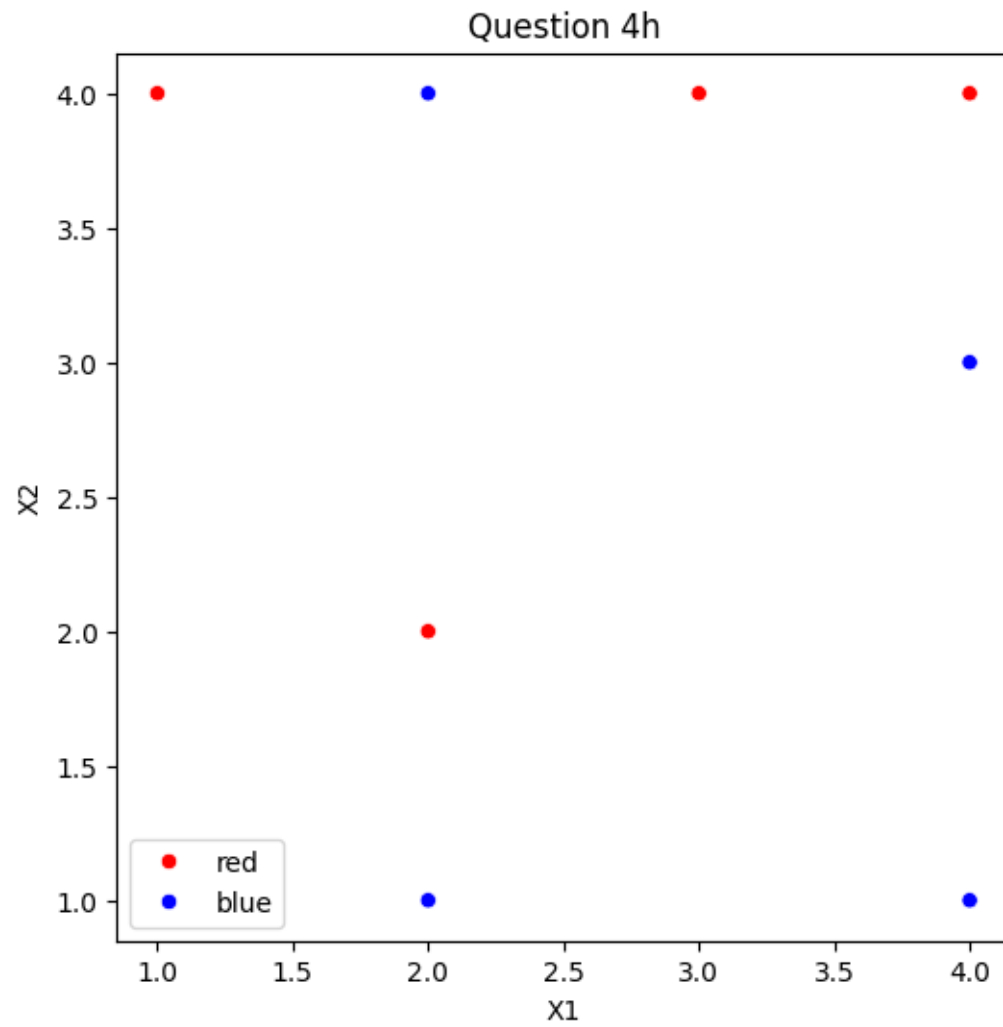
#### Question 4



- c. Classify to Red if  $\beta_0 + \beta_1 X_1 + \beta_2 X_2 > 0$  and Blue otherwise.  
In this case,  $\beta_0 = -0.5$ ,  $\beta_1 = 1$ ,  $\beta_2 = -1$
- f. The 7th observation (4,1) if moved slightly will not affect the hyperplane because it is not a support vector, the equation of the hyperplane does not depend on it.
- g. The equation for this hyperplane is  $0 = 0.8X_1 - X_2$



- h. A point that would make the two classes unseparable by a hyperplane would be (2,4) with the category Blue.

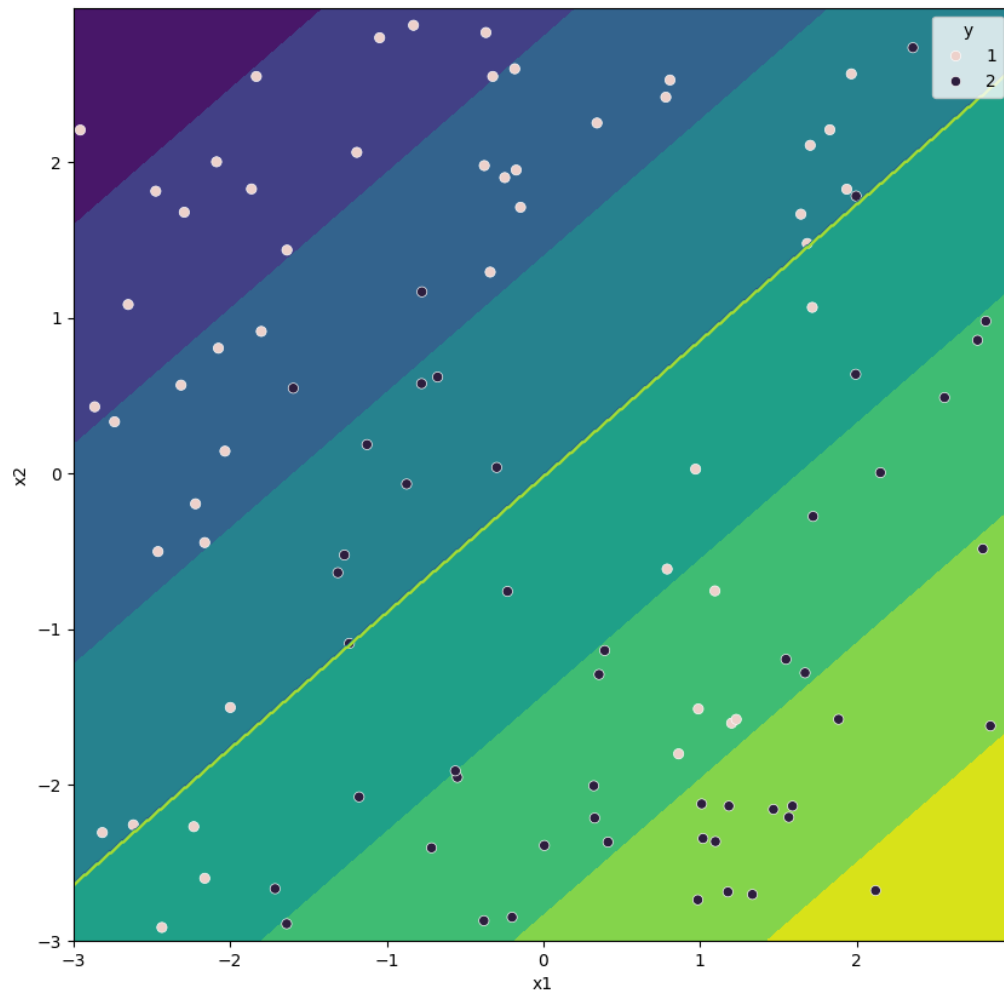


### Question 6

Linear kernel support vector machine

Training score: 0.77

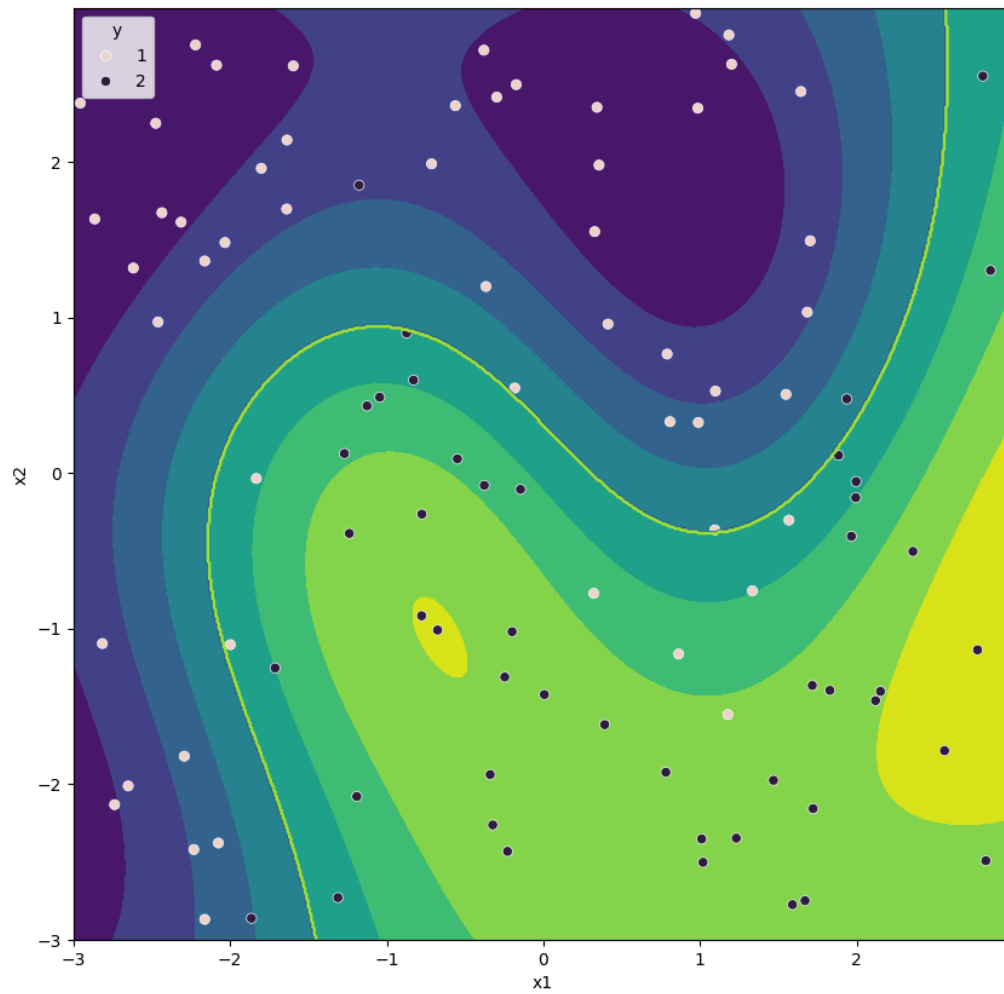
Test score: 0.72



Radial kernel support vector machine

Training score: 0.89

Test score: 0.79



Therefore, radial kernel works better in this case of non-linear separation between the two classes.